Lesson plan · for the teacher

Best choices: unit check

Optimization · Level 1 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- What makes a choice the best?
- The shortest way
- Quick rules and their traps
- The best for whom?

Activity: Decision detectives

1. Warm-up (5 min): Collect real school decisions on the board: the seating plan, the duty rota, the sports day order, the lunch queue.
2. Three parts (10 min): Each group takes one decision and writes its options, its limits and its goal in three columns.
3. Quick rule (10 min): Groups make a plan with one quick rule (such as first come, first served or nearest first) and check it keeps every limit.
4. Fair? (10 min): Groups find who comes off worst in their plan and suggest one change that would be fairer, even if the goal score drops a little.
5. Share (5 min): Each group shares its decision, its plan and one improvement. The class votes on the fairest idea.

Use made-up names if a real decision is sensitive. The aim is to practise naming options, limits and goals, and asking who pays.

Discussion

- Which idea from this unit will you use first?
- Which quick rule do we trust too much at our school?
- When is a slightly lower score worth it for fairness?

Exit ticket

- One decision I will plan differently, and how.

Try it in class tomorrow

Pick one decision your school makes this month (a rota, the trip, exam rooms). With a colleague, write its options, limits and goal, try one quick rule, and ask who the plan is hard on.

Worksheet · for pupils

Best choices: unit check

Optimization · Level 1 · Unit check

Name _____ Class _____ Date _____

1 **Amina's class has 1,200 coins and five outing ideas, and wants the most votes. What is the limit?**

◯ The five outing ideas ◯ The budget of 1,200 coins ◯ The most votes

2 **A feasible plan keeps every limit, so it is always the optimal plan.**

◯ True ◯ False

3 **Which of these are goals: one number to make as large or as small as possible? Choose all that apply.**

☐ The least fuel for the bus round ☐ No teacher in two classes at once

☐ The most votes for the outings ☐ At most 1,000 kg on the truck

☐ The shortest walk to the clinic

4 **Put in order what happens on the village map with Dijkstra's method.**

☐ The school is settled: the shortest way is 13 km

☐ Mina is settled, and the school gets 13

☐ Bora, Rubi and Kesi are settled, and Mina drops to 9

☐ Tala is settled, and Kesi gets 7

☐ Home gets 0; Tala gets 4, Bora 5 and Rubi 6

5 **The bridge between Kesi and Mina floods. What is the shortest way to school now?**

◯ Through Bora: 15 km ◯ Through Rubi and Sana: 14 km

◯ Through Tala, Kesi and Mina: 13 km